

SRI GANGANAGAR, India

WMO: 421230

Lat: 29.917N

Lon: 73.883E

Elev: 177

StdP: 99.22

Time Zone: 5.50 (IND)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.1	(c) 6.3	(d) 1.4	(e) 4.3	(f) 16.5	(g) 3.2	(h) 4.9	(i) 14.5	(j) 2.8	(k) 14.5	(l) 2.3	(m) 14.9	(n) 0.1	(o) N/A	(p) 0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 11.1	(c) 43.9	(d) 24.4	(e) 42.4	(f) 24.3	(g) 41.0	(h) 24.6	(i) 29.0	(j) 35.6	(k) 28.6	(l) 35.1	(m) 28.3	(n) 34.6	(o) 1.0	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.7	24.1	31.7	27.3	23.6	31.7	26.9	23.1	31.7	96.5	35.7	94.9	35.3	93.3	34.8	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.0	2.4	2.1	2.9	46.3	1.4	1.8	1.9	47.6	1.1	48.7	0.3	49.7	-0.7	51.1		
			2.7	29.7	1.2	0.5	1.8	30.1	1.1	30.4	0.4	30.6	-0.5	31.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	12.5	16.4	22.1	28.4	33.1	34.4	33.3	32.2	30.2	26.2	19.5	14.1
	DBStd	8.10	2.40	2.63	3.33	3.32	2.89	3.13	2.27	1.92	2.12	2.76	2.27	2.80
	HDD10.0	9	6	0	0	0	0	0	0	0	0	0	0	3
	HDD18.3	398	180	66	7	0	0	0	0	0	0	0	13	132
	CDD10.0	5574	84	179	374	552	715	731	723	688	607	503	286	131
	CDD18.3	2921	0	12	123	302	457	481	465	430	357	245	48	2
	CDH23.3	42938	1	101	1323	4203	7458	7931	7340	6453	4907	2783	420	18
(13)	CDH26.7	26893	0	13	548	2553	5137	5582	4880	3995	2726	1383	74	2
(14) Wind	WSAvg	0.5	0.3	0.5	0.5	0.6	0.7	0.8	0.7	0.6	0.5	0.2	0.1	0.2
(15) Precipitation	PrecAvg	331	8	17	12	12	17	36	76	71	58	8	2	3
	PrecMax	514	22	57	78	45	66	120	165	189	229	76	16	25
	PrecMin	156	0	0	0	0	0	5	1	1	0	0	0	0
	PrecStd	98	7	18	20	12	18	33	45	56	64	16	3	6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	28.4	36.7	42.7	45.9	46.1	42.9	39.7	39.1	37.2	30.5	25.4
		MCWB	15.3	18.4	20.8	21.6	23.3	24.1	27.2	27.2	25.7	23.2	19.8	15.8
	2%	DB	20.9	26.2	34.0	40.6	44.0	44.2	40.6	38.7	37.7	35.5	28.5	23.3
		MCWB	15.0	17.7	20.5	21.1	23.2	25.4	27.0	27.2	25.5	23.1	19.3	15.9
	5%	DB	19.5	24.2	31.7	38.9	42.4	42.7	39.4	37.7	36.7	34.3	27.2	21.9
		MCWB	13.8	16.7	20.3	20.7	23.1	25.6	27.2	27.0	25.5	23.2	19.0	15.3
	10%	DB	18.1	22.4	29.7	36.9	40.9	41.2	38.2	36.7	35.6	32.9	25.8	20.5
		MCWB	13.1	15.7	19.6	20.5	23.1	25.9	27.4	27.2	25.8	22.7	18.4	14.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	20.3	23.2	24.2	27.3	29.1	29.5	29.4	29.1	26.7	21.3	17.9
		MCDB	20.5	25.3	31.4	37.4	38.9	37.7	36.0	34.8	34.4	33.8	27.4	22.5
	2%	WB	15.6	18.9	22.1	22.8	26.0	28.5	29.0	28.9	28.4	25.7	20.5	16.7
		MCDB	19.7	24.0	30.6	35.2	38.8	37.1	36.0	34.5	33.4	32.2	26.8	21.7
	5%	WB	14.6	17.8	21.3	22.0	25.1	28.0	28.7	28.6	27.8	24.8	19.8	15.8
		MCDB	18.5	22.6	29.2	34.4	37.9	36.6	35.5	34.0	32.6	30.8	26.0	21.0
	10%	WB	13.7	16.7	20.4	21.4	24.3	27.5	28.3	28.2	27.2	23.9	19.0	14.8
		MCDB	17.1	21.2	28.1	33.5	37.0	36.1	35.0	33.5	32.0	30.3	24.6	19.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	11.2	12.7	14.7	14.0	11.1	8.6	7.7	9.2	12.6	11.8	11.1
		MCDBR	12.0	13.6	15.8	17.1	16.1	13.7	9.9	8.9	11.0	13.5	13.7	13.2
		MCWBR	6.5	6.1	5.1	4.3	4.0	3.3	2.5	2.4	2.7	3.9	5.9	6.7
	5% WB	MCDBR	10.4	11.6	13.9	14.6	13.3	10.1	8.8	7.8	8.2	10.6	11.5	11.9
		MCWBR	6.3	5.8	5.5	4.4	4.3	3.4	2.7	2.5	2.9	3.6	5.6	6.5
(40) Clear-Sky Solar Irradiance	taub		0.490	0.476	0.503	0.580	0.668	0.761	0.793	0.686	0.592	0.634	0.608	0.533
	taud		1.838	1.928	1.889	1.751	1.608	1.493	1.474	1.660	1.800	1.618	1.629	1.750
	Ebn at Noon		706	774	785	741	679	616	594	655	702	622	589	634
	Edh at Noon		175	174	193	229	265	296	301	248	208	235	215	183
(44) All-Sky Solar Radiation	RadAvg		3.04	4.21	5.52	6.46	6.91	6.29	5.70	5.65	5.45	4.65	3.54	3.05
	RadStd		0.32	0.30	0.27	0.25	0.18	0.34	0.31	0.34	0.25	0.22	0.22	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page